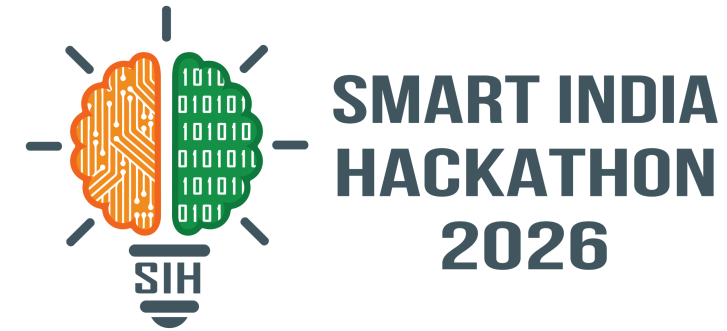


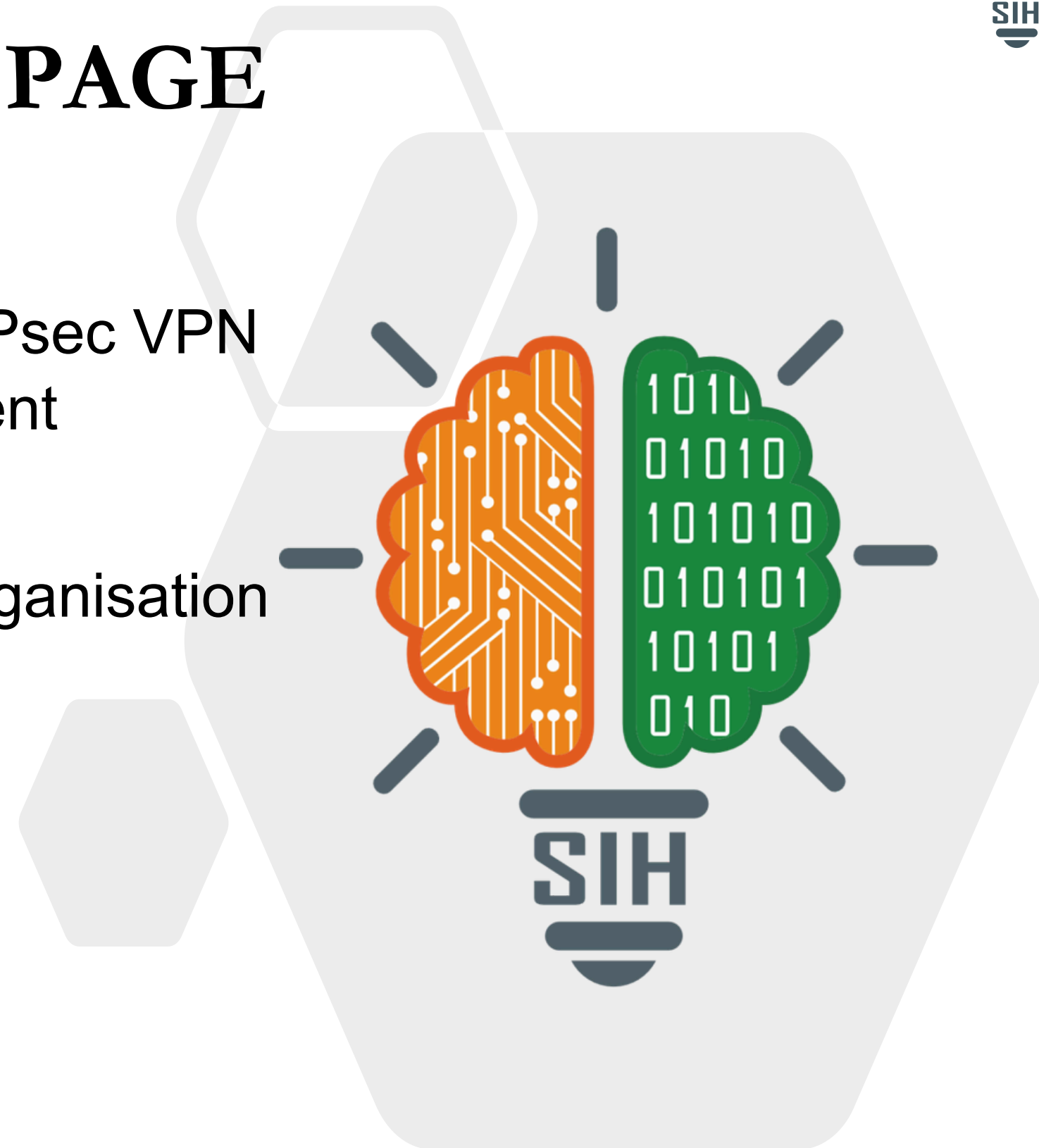


SMART INDIA HACKATHON 2026



TITLE PAGE

- **Problem Statement ID** - 26160
- **Problem Statement Title** - AI-Powered IPsec VPN Protocol Analyzer and Security Assessment Framework
- **Theme** - National Technical Research Organisation (NTRO)
- **PS Category** - Software
- **Team ID** -
- **Team Name** - The Alchemist



AI-POWERED IPsec VPN PROTOCOL ANALYZER AND SECURITY FRAMEWORK



From raw VPN traffic to explainable, evidence driven security assessment

Core problems Faced:

Manual Packet Analysis

Analysts must inspect complex VPN traffic one packet at a time.

Encrypted Traffic Is Hard to Understand

ESP payloads cannot simply be read or decrypted.

Uncertain AI Can Mislead Analysts

Predictions may be confused with confirmed protocol facts.

Security Gaps Can Be Missed

Unsafe configurations and weak security posture may go unnoticed.

Raw Results Are Difficult to Act On

Technical packet data is not easy for decision-makers to use.

How we tackle them:

Automated Protocol Detection

Detects observable IKE, ESP, AH and NAT-T traffic automatically.

Privacy-Preserving AI

Uses packet size, timing, direction and burst patterns without decrypting ESP.

Confidence-Aware Results

Shows confidence and returns UNKNOWN when evidence is insufficient.

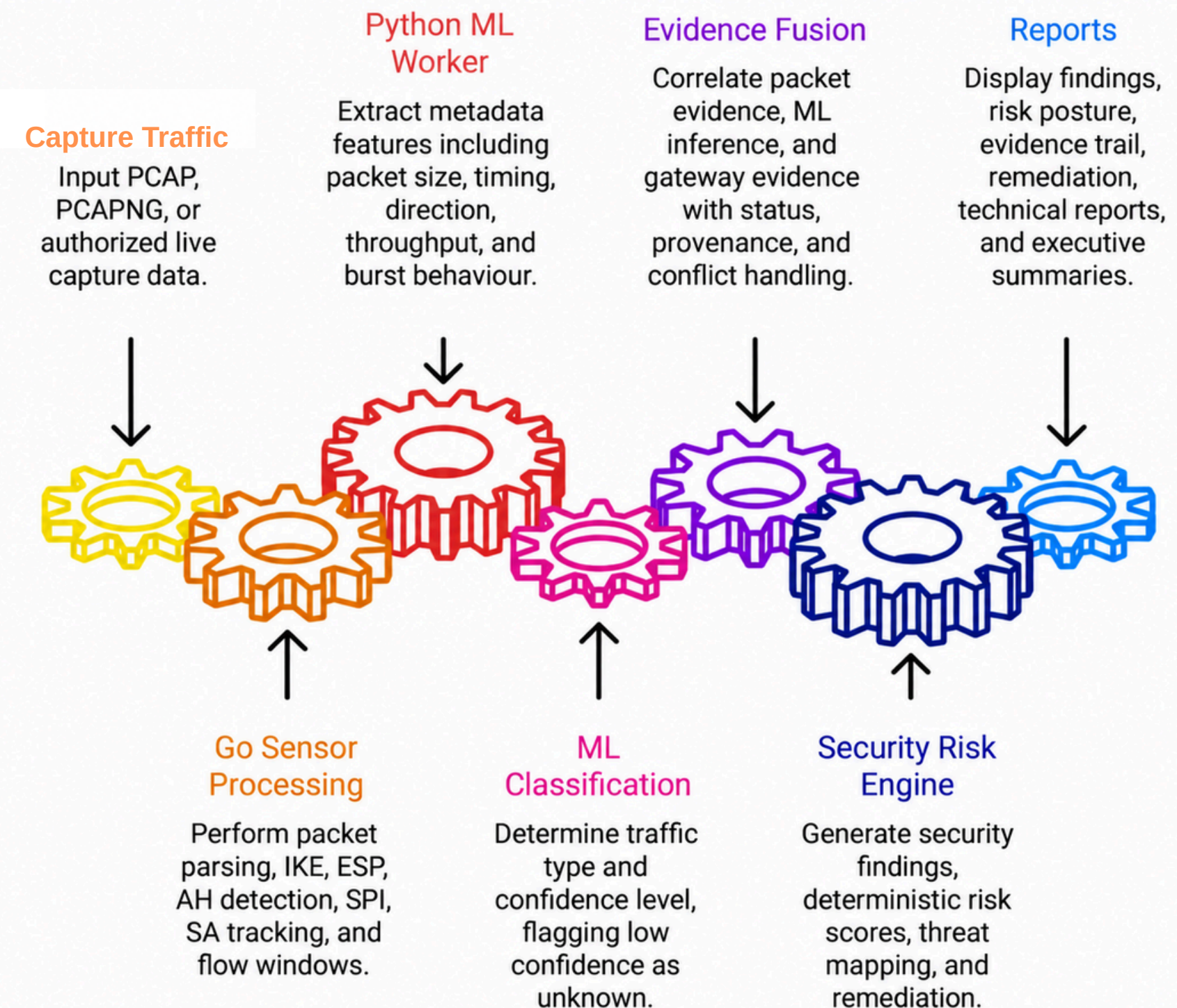
Rule-Based Security Assessment

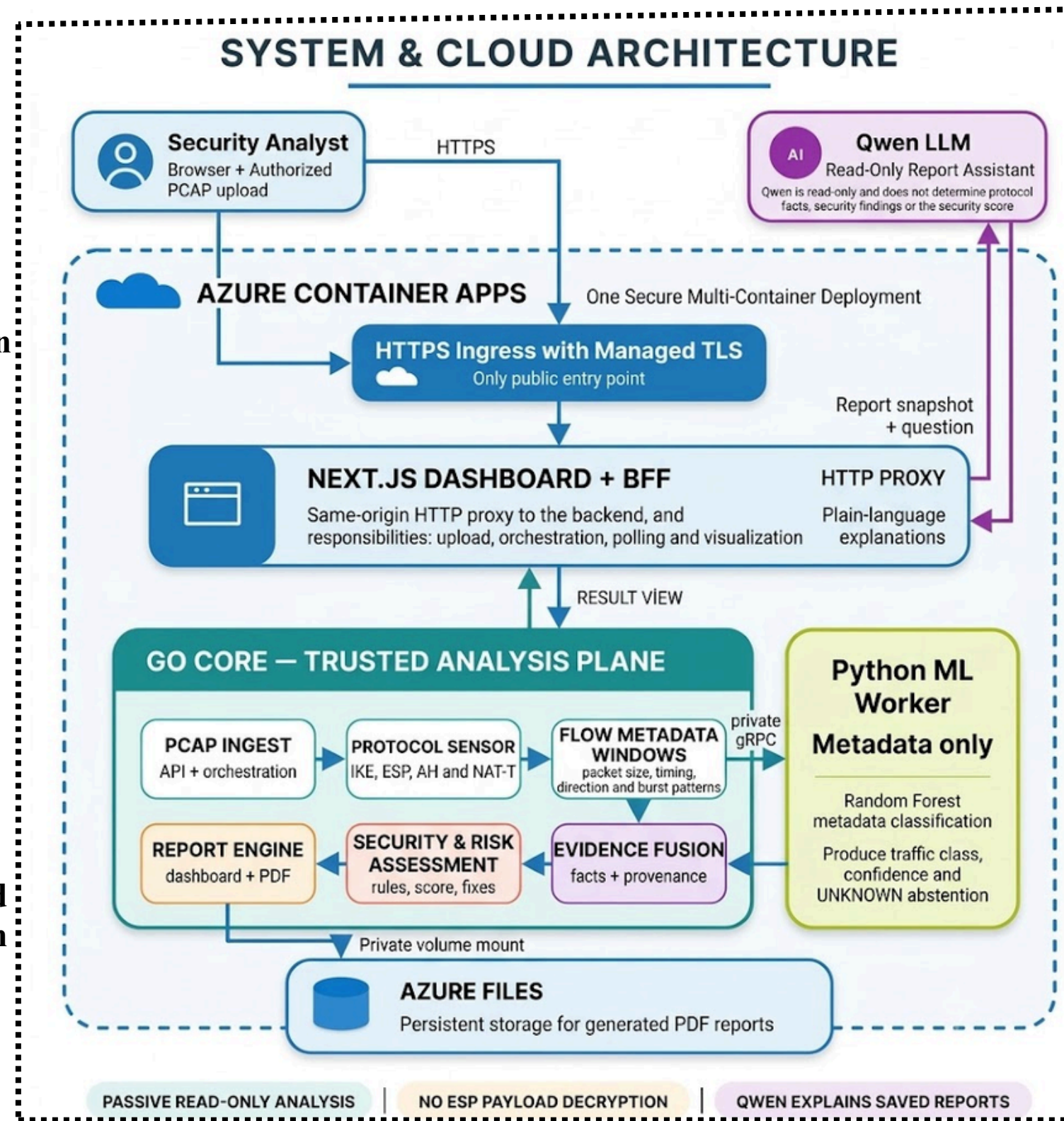
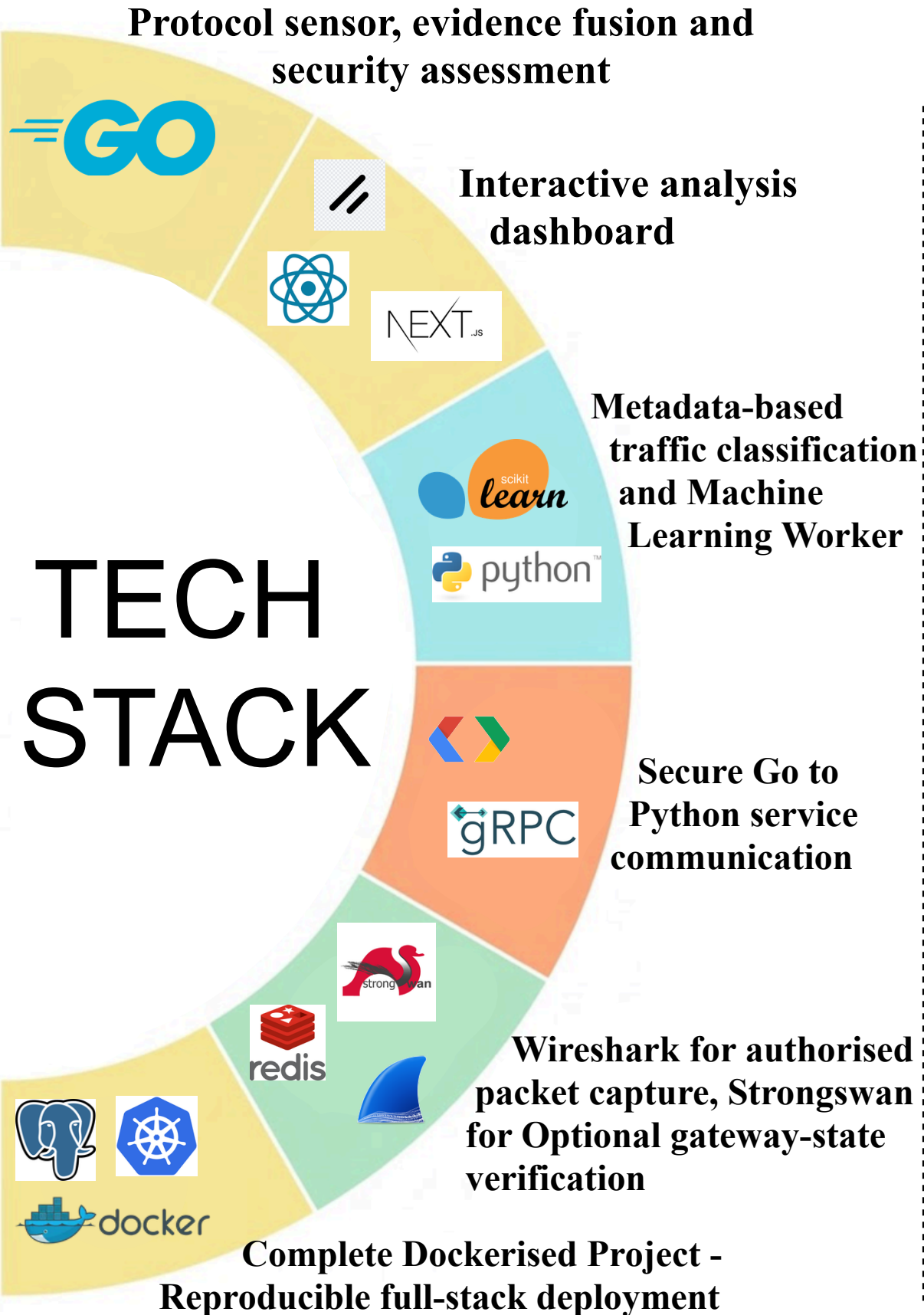
Creates evidence-backed findings, risk level and remediation steps.

Dashboard & PDF Reports

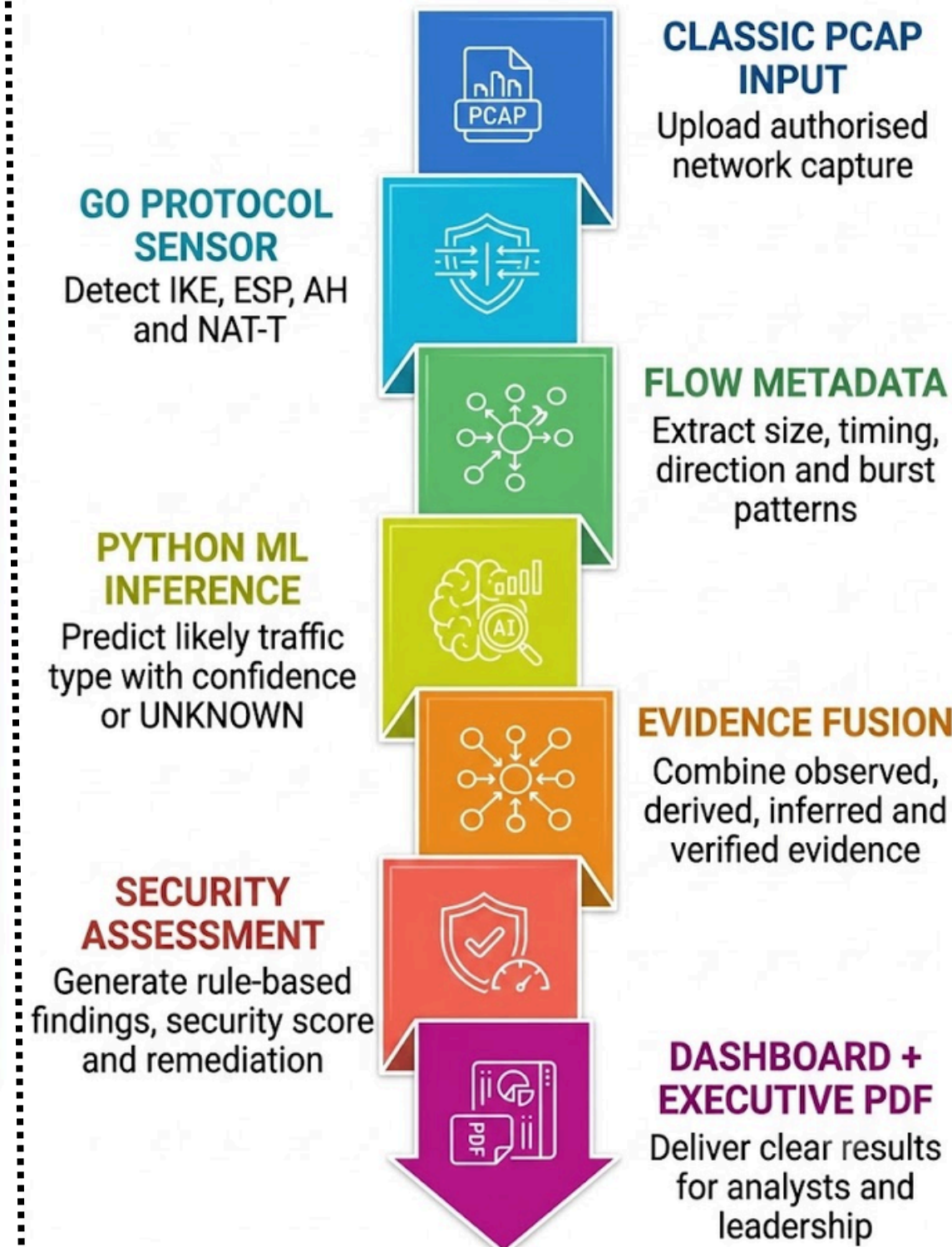
Converts analysis into clear results for analysts and leadership.

FRAMEWORK





Methodology and Process

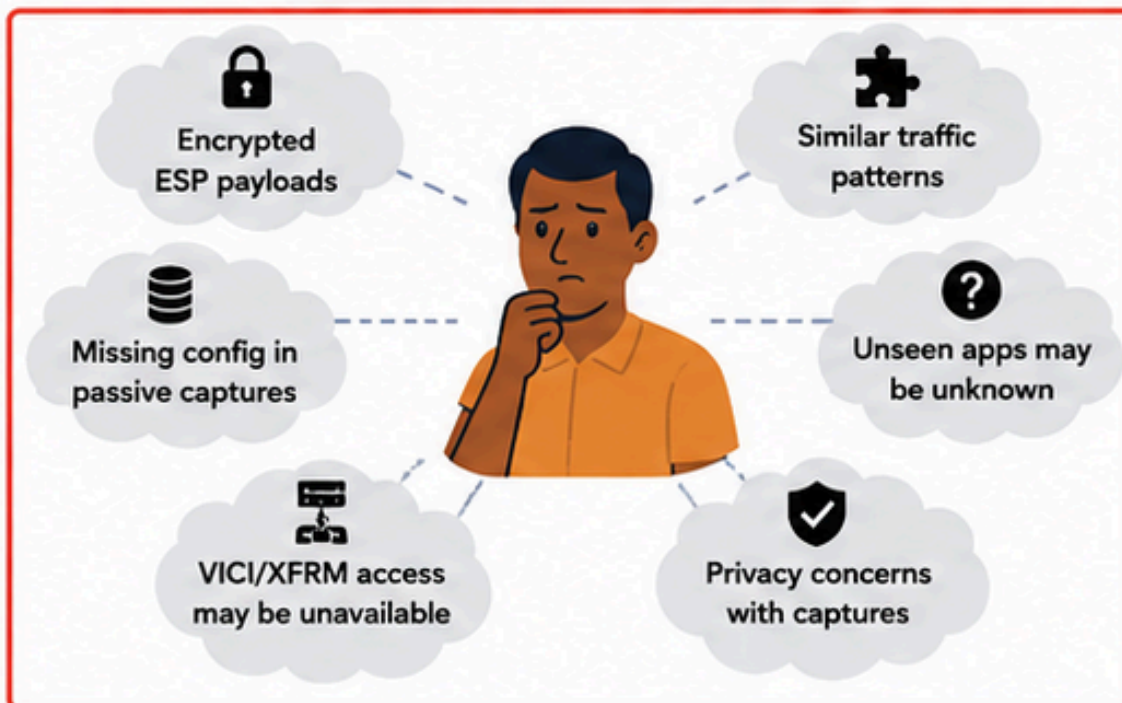


FEASIBILITY

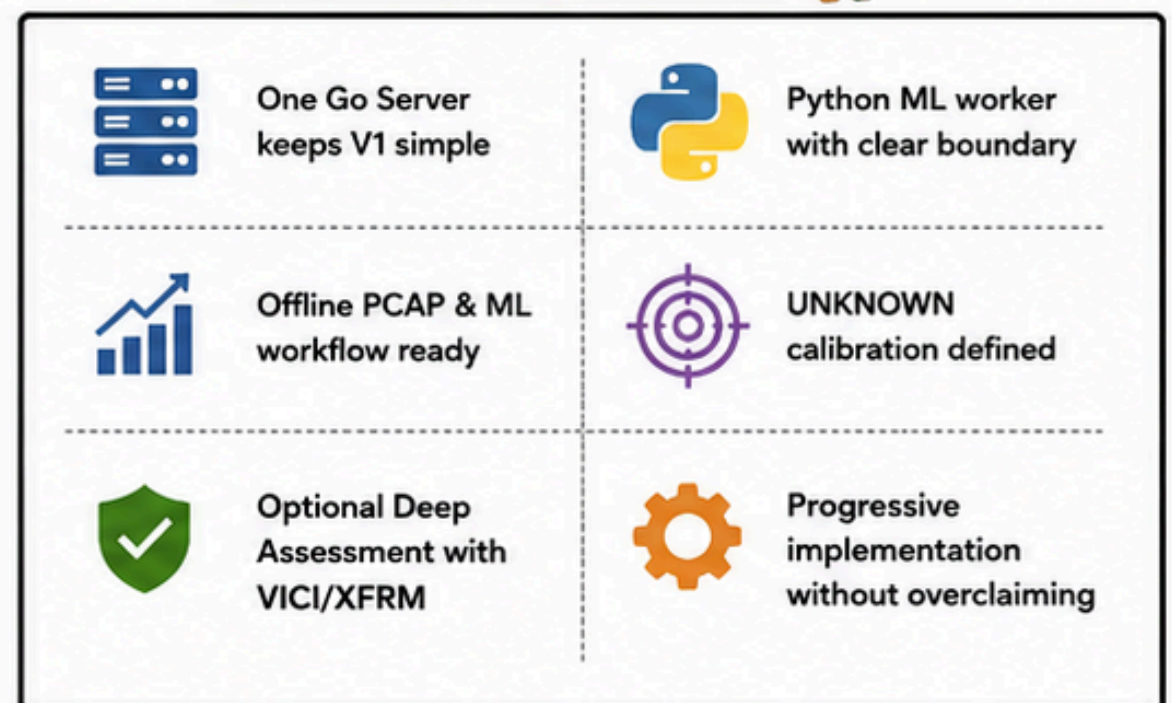
- **One Go Server architecture** keeps V1 deployment **simple** and easy to maintain.
- **Python ML worker** has a clear, **typed process boundary** receiving only flow metadata.
- **Offline PCAP metadata** extraction & ML workflow are available.
- **UNKNOWN calibration** and anomaly detection workflows are defined.
- **Deep Assessment** with **VICI/XFRM** provides optional gateway verification.

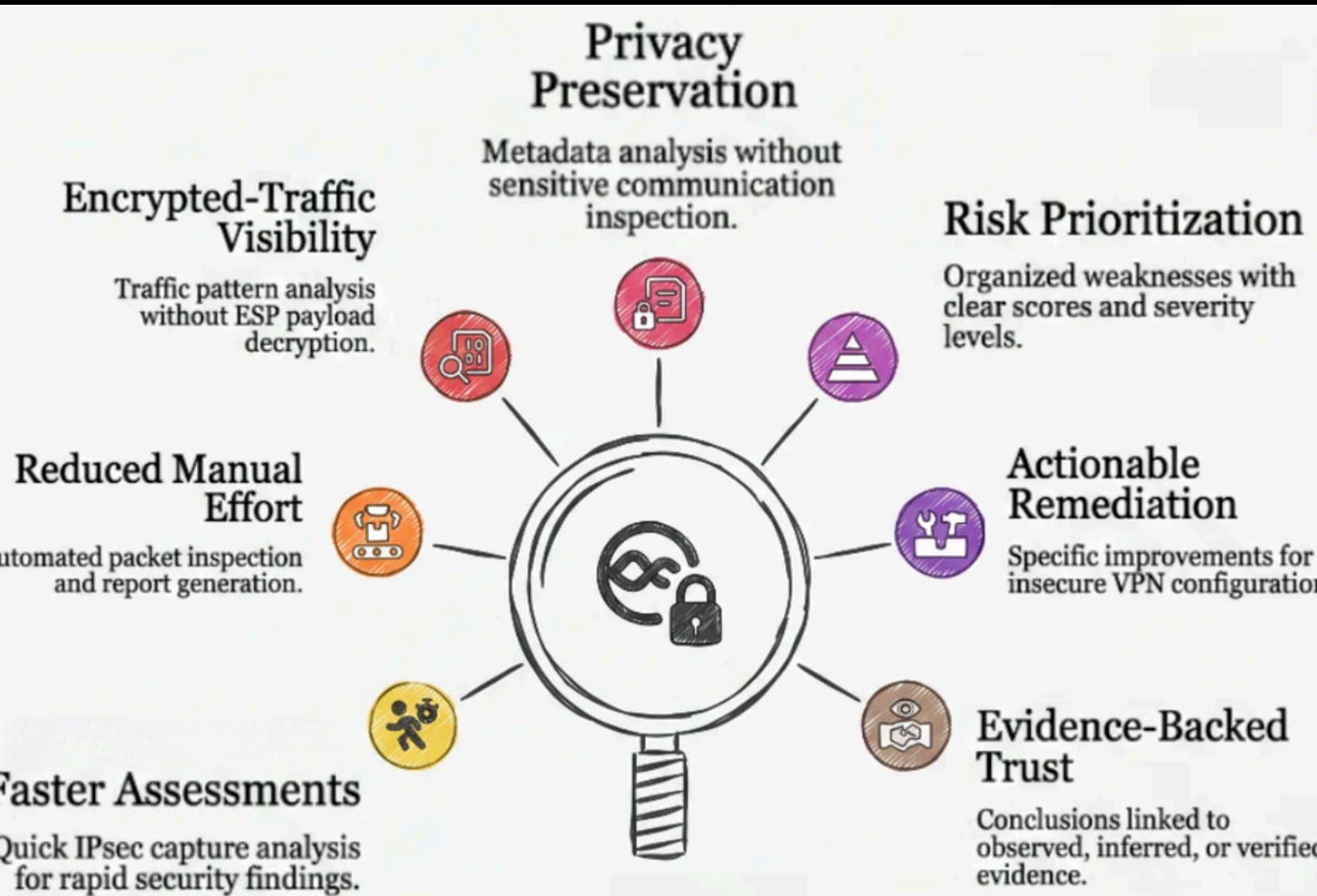
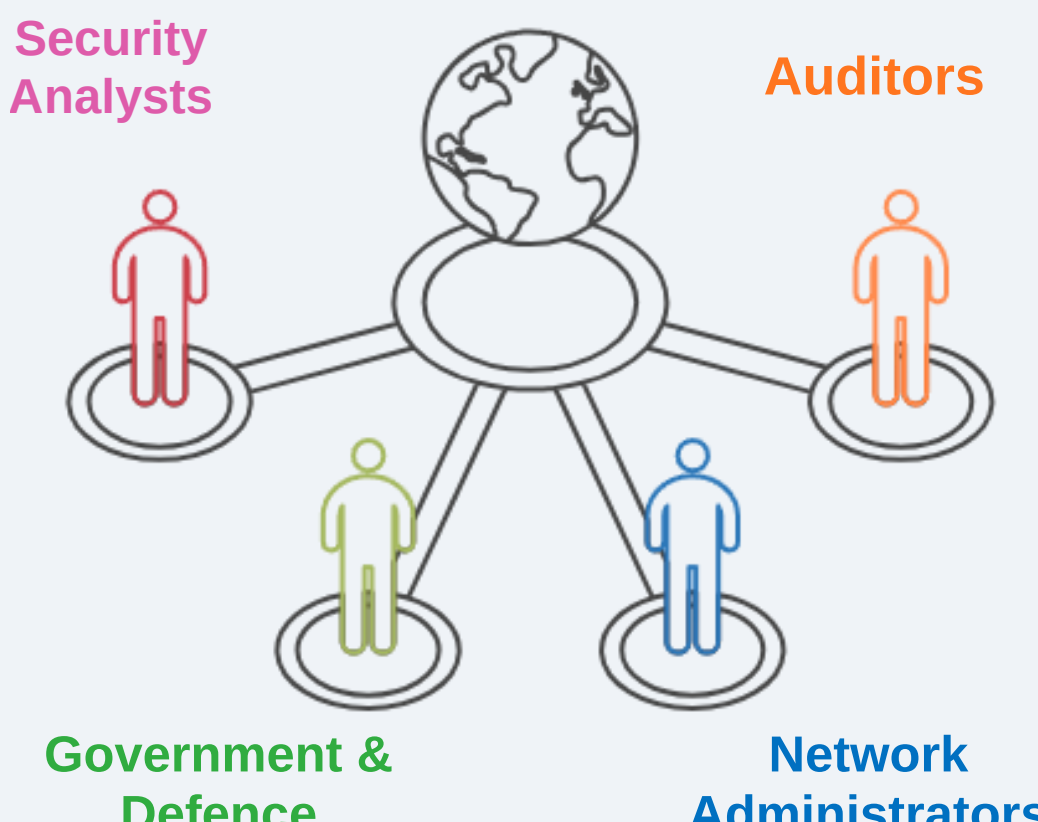
**VIABILITY**

- **Evidence-Driven** - Clear separation of observed, derived, inferred, verified & unknown evidence **builds trust**.
- **Risk Reduction** - Actionable findings and deterministic risk scores **reduce security exposure**.
- **Operational Efficiency** - Automation and smart analytics **reduce manual effort and cost**.
- **Scalable & Resilient** - Modular design supports **graceful fallbacks** and future enhancements.
- **Policy Alignment** - Supports Digital India, Make in India and secure, **local-first analysis**.

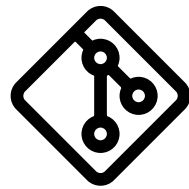
**POTENTIAL CHALLENGES****STRATEGIES TO OVERCOME**

- ✓ Use only **observable flow metadata**; never decrypt ESP payloads.
- ✓ Separate evidence into **observed, derived, inferred, verified** and **unknown**.
- ✓ Calibrated **UNKNOWN rejection** for low-confidence traffic.
- ✓ Keep packet/protocol facts **separate** from ML predictions.
- ✓ Use optional **VICI/XFRM** telemetry for authorized gateway verification.
- ✓ **Gracefully continue** analysis if ML, VICI or XFRM is unavailable.

FEASIBILITY HIGHLIGHTS

★ BENEFITS		★ BROADER BENEFITS	👥 TARGET AUDIENCE
		 Social Benefits - Stronger Digital Trust - Privacy Protection - Cybersecurity Accessibility   Economic Benefits - Lower Assessment Costs - Faster Risk Mitigation - Improved Productivity   Environmental Benefits - Reduced Resource Usage - Lower Travel & Paper Use - Longer Infrastructure Life   SUSTAINABLE AND SECURE DIGITAL ECOSYSTEM Safer Networks for a Better Tomorrow	
 OVERALL IMPACTS	Faster Assessments Accelerates security assessments for quicker, more efficient results.	Reduced Analyst Effort Automates repetitive tasks and reduces analysts' manual workload.	Better Visibility Provides clearer insights into risks and overall security posture.

PROJECT LINKS

[Demo Video](#)[Source Code](#)[Deployed Link](#)

REFERENCES

- **IEEE – Encrypted Traffic Analysis and Classification (2008)**
Machine-learning based analysis of encrypted VPN traffic.
<https://ieeexplore.ieee.org/document/4564875>
- **IEEE – VPN Traffic Characterization (2012)**
Identification and characterization of VPN traffic patterns.
<https://ieeexplore.ieee.org/document/6168355>
- **NIST SP 800-77 Rev.1**
Guide to IPsec VPN deployment, security recommendations and best practices.
<https://nvlpubs.nist.gov/nistpubs/SpecialPublications/NIST.SP.800-77r1.pdf>

DATASET AND RESEARCH INVESTIGATION

- **IPsec PCAP Lab (GitHub) [Team-Generated Dataset]**
Collection of IPsec traffic captures and test scenarios.
<https://github.com/naman9271/ipsec-pcap-lab>
- **MIT VPN/Non-VPN Network Application Traffic Dataset**
Benchmark dataset for VPN and non-VPN traffic classification.
<https://www.ll.mit.edu/r-d/datasets/vpnnonvpn-network-application-traffic-dataset-vnat>
- **Wireshark Sample Captures**
Reference ESP payload decryption and authentication checking examples.
<https://wiki.wireshark.org/samplecaptures#example-1-esp-payload-decryption-and-authentication-checking-examples>

